

MEASUREMENT OF CLUSTERING TENDENCY*

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Abstract. Determining the structure of multi-dimensional data is an important problem in exploratory data analysis and pattern recognition. Clustering methods have been used extensively for this purpose. However, clustering algorithms will locate and specify clusters in data even if none are present. It is therefore appropriate to measure the clustering tendency or randomness of a data set before subjecting it to a clustering algorithm. Hopkins' method of testing for randomness is extended to high dimensions and is tested against data from clustered and hardcore processes along with the Fisher Iris data. As in two dimensions, it appears to be a powerful test for clustering tendency.

Keywords: Cluster Analysis, Data Reduction and Analysis, Pattern Recognition, Probability, Statistics, Hopkins' Method.

INTRODUCTION

Given a sample of N patterns $X(1), X(2), \dots, X(N)$ in d -dimensional Euclidean space, we want to determine if there is a tendency for these patterns to fall into natural disjoint groups, called clusters. A wide collection of clustering algorithms is available, each operating under different criteria [1,9,11,8.] The issue of cluster validity has virtually been ignored in the literature and this makes it difficult to interpret the results of a clustering algorithm [7]. Since most clustering algorithms will always identify some clusters in data, there is a need to develop tests to determine if data exhibits clustering before actually applying any clustering algorithms. The availability of such tests will strengthen the statistical foundations of clustering methodology. Unfortunately, this problem has received very little attention in pattern recognition and exploratory data analysis.

Data can be presented in two ways to a clustering algorithm. The first data type is called the proximity matrix, in which we do not have the actual coordinates of the patterns but only a measure of similarity or dissimilarity between the patterns. This type of data occurs frequently in social and behavioral sciences. Some clustering tendency research has been done on proximity data by means of random graph theory applied to ordinal proximity matrices [7]. The other data type, which is of interest here, is the pattern matrix, which consists of a number of measure-

ments made on each pattern. A number of techniques for the analysis and modeling of two-dimensional patterns is available in the literature [5,21,22,23]. The development of these techniques was motivated by a need to interpret and analyze large amounts of two-dimensional data collected in various ecological and socio-geographic studies. These tests have not been tried out on high-dimensional data and it is not known whether a direct generalization (to dimension $d > 2$) of these statistics would lead to useful tests. The main purpose of this study is to see whether some of the test statistics known to be powerful in two-dimensions can be generalized to handle high-dimensional data.

We wish to pose the clustering tendency problem in a hypothesis testing framework. Essentially, we want to describe a set of patterns as being "random," "clustered," or "regular." In some sense, we are trying to determine the gross structure of the multi-dimensional data. Our null hypothesis will always be that the given data is "random." If the null hypothesis fails to be rejected at some significance level then we will say that the data does not exhibit a tendency to cluster.

The next section introduces spatial point processes which are used to give a formal definition of our concept of "random," "clustered," and "regular" structure. This is followed by the description of a class of powerful tests for analyzing two-dimensional data which also appears to be very promising in assessing the randomness of high-dimensional data. The details of experiments

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DISTANCE METHODS

Distance-based tests of randomness use statistics compiled from near neighbor distances of an arbitrary point of space to a point in the given realization of the underlying process. Our focus will be on the adaptation of the known two-dimensional methods to high-dimensional data ($d > 2$).

One may wonder why we do not simply compute the histogram of near neighbor distances and then compare it to the theoretical one under the Poisson hypothesis or other alternative. This can be done using the Kolmogoroff-Smirnov test or Chi-square goodness of fit test. Unfortunately, such a method would have little power. Consider pairs of points whose distances from each other obeys the law associated with a dense Poisson field, but let the pairs be dispersed far from each other. In this case, either of the two tests would find that the first near neighbor distribution matches, but the field is certainly not Poisson. To counter this one could compare the empirical joint distribution of the first k near neighbor distances to that expected under the Poisson null distribution. However, fitting a multivariate density is a notoriously difficult problem and the available methods such as the extension of the Kolmogoroff-Smirnov test or Chi-square test are of low power also.

Hence, all the distance-based methods use some sampling 'points' in addition to the given data 'events.' These M points are placed in the previously identified sampling frame according to a Poisson distribution, which is to say uniformly. The sampling points are few enough in number, relative to N , the number of events that their presence does not materially affect the overall density. Ratios of at least ten to one and preferably twenty to one are used in the literature. On the other hand, it seems that M should be at least 10 in order to avoid any small sample problems with the distributions of the statistics. This effectively limits the methods to problems with at least 100 events. In high dimensions, very little can be said about data sets that are sparser than that.

Heterogeneity of the data, whereby vast areas of the sampling frame contain no pattern and other areas contain large numbers in tight clusters, can be detected by the use of these sampling points. The heterogeneity will show up in that large distances will be observed in the case of the first near neighbor of a point to an event. Such observations would be inconsistent with the presence of a Poisson field and would lead to the rejection of the Poisson null hypothesis. We now give the details of one of the most powerful distance-based methods, Hopkins' test. Diggle et al. [6] perform a comparison of the relative strength of Hopkins' test with other distance methods and find that it was the strongest among four other alternatives. Hopkins' [15] method for assessing random-

ness is one of the earliest methods and involves the distribution of nearest neighbor distances in a Poisson process. It is easy to derive that the distribution of nearest neighbor distance in a Poisson process of intensity λ is given by:

$$(1) f(r) = \lambda r^{d-1} \exp(-\lambda r^d V)$$

where V is the volume of a d -dimensional sphere of radius 1.

Let $\{Y(1), Y(2), \dots, Y(M)\}$ be M points placed at random in the sampling frame. Let $U(i)$ be the distance from $Y(i)$ to the closest $X(j)$ for $i = 1, 2, \dots, M$; $j = 1, 2, \dots, N$. We assume that $M \ll N$. Let $\{W(i)\}$, $i = 1, 2, \dots, M$ be a random sample of M near neighbor distances among the X 's. Under the null hypothesis of Poisson, $\{U(i)\}$ and $\{W(i)\}$ have an identical distribution given by equation (1). Hopkins' test is based on the fact that the random variable

$$(2) H = \sum_{i=1}^M \frac{U(i)^d}{U(i)^d + W(i)^d}$$

has a Beta distribution with parameter (M, M) .

EXPERIMENTAL RESULTS

Three types of data were used to test the viability of Hopkins' method: Hardcore, Clustered, and Iris data. Poisson is a special case of Hardcore. All testing was performed on a PDP 11/34 minicomputer under RT-11 or an IBM 3033 under MVS.

HARDCORE MODEL RESULTS

Data was generated according to the hardcore model using the simulation procedure mentioned above. This is the so-called SSI process of Diggle [6]. The key parameter is ρ , the proportion of the sampling frame volume covered by the hardspheres. When ρ is 0.00, there is no inhibition and the data is Poisson or random. A maximum covering proportion of 0.9765 is possible. The number of points was kept uniformly to be 200, and 100 realizations at each value of ρ were created. The data was generated in the unit hypercube using a torus topology, i.e. distances between points are computed using a periodic boundary condition. The number of rejections of the null hypothesis of spatial randomness at the level of 0.05, two-tailed is shown in Table 1 at each value of ρ and each dimension, d . Hopkins' test rejects the hypothesis of randomness even at rather low values of the parameter for $d > 2$ and $d = 2$.

CLUSTERED DATA

Clustered data was generated in the unit hypercube according to the Neyman-Scott model described above. There are two parameters: μ , the intensity of the Daughter process, and σ , the standard deviation of the multivariate normal radial distribution function. A

utilizing a distance-based method are then given, followed by the conclusions of our investigation.

SPATIAL POINT PROCESSES

Our framework for studying the clustering tendency problem is based on spatial point processes, and follows the work of Hammersley [10] and Ripley [22]. Spatial point processes are the analogs of one-dimensional stochastic processes. They generalize the essential features of the defining properties in one dimension, usually replacing an interval by a bounded set. The random variable of interest is $Z(A)$, the number of points of the process in a bounded measurable set A . The collection $\{Z(A)\}$ as A ranges over the Borel sets constitutes the process. The types of spatial processes are distinguished by the distributions of $Z(A)$ and the correlations between them. We assume for any realization of the process, that there are no limit points and, *a fortiori* that $Z(A)$ is finite for all bounded A . We must point out that Bartlett [2] has a different point of view of a spatial point process than taken here.

POISSON PROCESS

The simplest and most important case of spatial point process is Poisson process. The Poisson process is defined by two conditions. First, $Z(A)$ is a discrete integer valued Poisson variable with intensity parameter $\lambda v(A)$, where $v(A)$ denotes the measure of A . The quantity λ is called the intensity of the process. The second condition is that if $\{A\}$ is a collection of disjoint sets of arbitrary cardinality, then $\{Z(A)\}$ is a set of independent random variables. One way to simulate a Poisson process is to specify a set A and place N points uniformly in A , where N is distributed as a Poisson variable with parameter $\lambda v(A)$. The resulting density is uniform, i.e. $p(X) = 1/v(A)$ for each point X in the set A . Figure 1 shows the realization of a Poisson process with parameter = 200. Our concept of randomness of multi-dimensional data is based on this definition of Poisson process. In fact, our tests of clustering tendency will involve testing the null hypothesis that the given data is a sample from a Poisson process.

CLUSTERED PROCESS

Among a large number of models for clustered processes, the Neyman-Scott process is perhaps the most well known. A Neyman-Scott process [19,20] consists of a parent process and a daughter process. The parent process is a spatial Poisson process which generates the cluster centers. The clusters themselves are generated by the daughter process which consists of a radial distribution function, usually taken to be multivariate Gaussian, and an integer Poisson distribution which controls the number of points per cluster. Figure 2 is a realization of a Neyman-Scott process.

Classical cluster analysis can thus be viewed as an attempt to estimate the position of the cluster centers and identify the members of the daughter process. The enormity of the problem and its overall intractability is further compounded if we lack any information on the number of cluster centers and the cluster size. In terms of statistical inference, this estimation is being done from a single sample, i.e., from a single realization.

HARDCORE PROCESS

The next general class of models is the so-called hardcore or inhibitory models. Whereas the clustered models have interpoint distance distributions which are below that expected by a Poisson process, these models all have the feature that no two points of the process are within a distance $2r$ of each other, where r is a fixed positive number called the interaction radius. In some sense, hardcore models represent a regular structure which we may characterize as anti-clustering or repulsion among patterns. They are frequently called hard-sphere models in the applied physics literature [4,24] since we can model them as the centers of spheres which do not overlap. The hardcore model has wide applicability to physical situations [16,18].

There are a number of models for generating hardcore processes. The details appear in Matern [18] and Ripley [22]. In general, they involve embedding Poisson process in a temporal framework. Points are dropped on a d -dimensional space and a point is retained if it is not within a distance $2r$ of any other point of the process. Another version simply chooses realizations of the Poisson process and retains only those realizations which obey the hardcore property (this is usually done only on a bounded set). The appearance of a hardcore realization is similar to a noisy lattice as shown in Figure 3. In fact, the lattice model is often used as an alternative to the Poisson model in the calculation of the theoretical power of tests of randomness.

The choice of a sampling window or bounded set over which the patterns are assumed to be dispersed is critical. In a data collection problem, the sampling window is usually imposed by the environment. For example, if we are studying the spatial distribution of trees, then the forest boundary specifies the sampling window. On the other hand, if we are simply presented with a set of pattern vectors, then these vectors are bounded by a hypercube and also by a hypersphere. The convex hull of the data will probably occupy only a small portion of the volume of the enclosing hypersphere or hypercube. The choice of a hypercube as a sampling window may make the data appear to be a single tight cluster in a large box and distort the structure of any micro-clusters in the data itself. In this paper, we will use a rectangular sampling frame.

total of 200 points were generated for each realization of the process. The results are shown in Table 2 (Dimension 2), Table 3 (Dimension 3), Table 4 (Dimension 4), and Table 5 (Dimension 5). As σ increases, the process is indistinguishable from a Poisson process, accounting for the low power. A similar result was obtained for the case of small μ , which means that each cluster has at most one member.

IRIS DATA

Hopkins' method was used to test the randomness of the Fisher Iris data. This data is four-dimensional and consists of 150 patterns. A preliminary transformation is necessary to apply Hopkins' method since the sampling frame is unknown. The sequence of steps necessary to standardize the data is:

1. Scale each feature to have mean zero.
2. Compute the eigenvalues and eigenvectors of the co-variance matrix.
3. Rotate the coordinate system to the principal axes defined by the eigenvectors.
4. Apply a linear transformation to each of the rotated features so that its range is $[0,1]$, with a 5 percent border at the edges.

Hopkins' method applied to this transformed Iris data yielded a value of the statistic H of .939717, which results in a rejection of the null hypothesis at the 0.05 level since the critical value for H is .563.

CONCLUSIONS AND DISCUSSION

The notion of clustering tendency was put into the framework of spatial point processes. The class of distance methods was discussed as a possible technique for measuring clustering tendency. Hopkins' method was extended to multi-dimensional data and was found to be powerful against both hardcore and clustered alternatives.

Further study is needed of the following:

1. Techniques for determining the sampling window.
2. Investigation of other distance methods such as those discussed by Diggle [6].
3. The influence of both the number of events and the number of sampling points on the distance method statistics.
4. The extension of the work of Ripley [22] to high dimensions.

REFERENCES

- [1] Anderberg, M.R., Cluster Analysis for Applications, New York: Academic Press, 1973.
- [2] Bartlett, M.S., The Statistical Analysis of Spatial Pattern, London: Chapman and Hall, 1976.
- [3] Besag, J.E., and Gleaves, J.T., "On the Detection of Spatial Pattern in Plant Communities," *Bull. Int. Stat. Inst.*, v45, pp. 153-158, 1958.
- [4] Croxton, C., Liquid State Physics, Cambridge: Cambridge University Press, 1974.
- [5] Cross, G.R., "Some Approaches to Measuring Clustering Tendency," Technical Report TR-80-03, Department of Computer Science, Michigan State University, 1980.
- [6] Diggle, P.J., Besag, J., and Gleaves, J.T., "Statistical Analysis of Spatial Processes by Means of Distance Methods," *Biometrics*, v32, pp. 659-667, 1976.
- [7] Dubes, R., and Jain, A.K., "Validity Studies in Clustering Methodologies," *Pattern Recognition*, v11, pp. 235-254, 1979.
- [8] Dubes, R., and Jain, A.K., "Clustering Methodologies in Exploratory Data Analysis," in Advances in Computers, v19, pp. 113-228, New York: Academic Press, 1980.
- [9] Everitt, B., Cluster Analysis, New York: John Wiley and Sons, 1974.
- [10] Hammersley, J.M., "Stochastic Models for the Distribution of Particles in Space," *Suppl. Adv. Appl. Prob.*, pp. 47-68, 1972.
- [11] Hartigan, J.A., Clustering Algorithms, New York: John Wiley and Sons, 1975.
- [12] Holgate, P., "Tests of Randomness based on Distance Methods," *Biometrika*, v52, pp. 345-353, 1965.
- [13] Holgate, P., "Some New Tests of Randomness," *J. Ecol.*, v53, pp. 261-266, 1965.
- [14] Holgate, P., "The Use of Distance Methods for the Analysis of Spatial Distribution of Points," in Stochastic Point Processes, New York: John Wiley and Sons, pp. 125-135, 1972.
- [15] Hopkins, B., with an appendix by J. Skellam, "A New Method for Determining the Type of Distribution of Plant Individuals," *Ann. Bot.*, v18, pp. 213-227, 1954.
- [16] Kamel, M.T., Tory, E.M., and Jodrey, W.S., "The Distribution of the k th Nearest Neighbours and its Application to Cluster Settling in Dispersions of Equal Spheres," *Powder Tech.*, v24, pp. 19-34, 1979.
- [17] Kendall, M.G. and Moran, P.A.P., Geometrical Probability, New York: Hafner, 1963.

FIGURE 1: Poisson Process

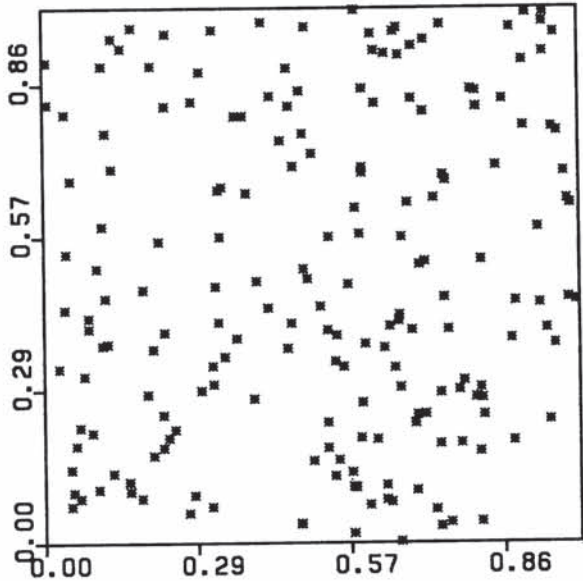


FIGURE 2: Cluster Process

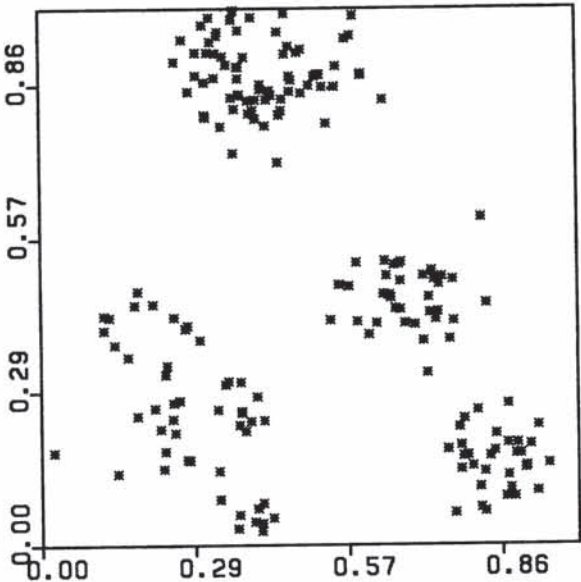


FIGURE 3: Hardcore Process

